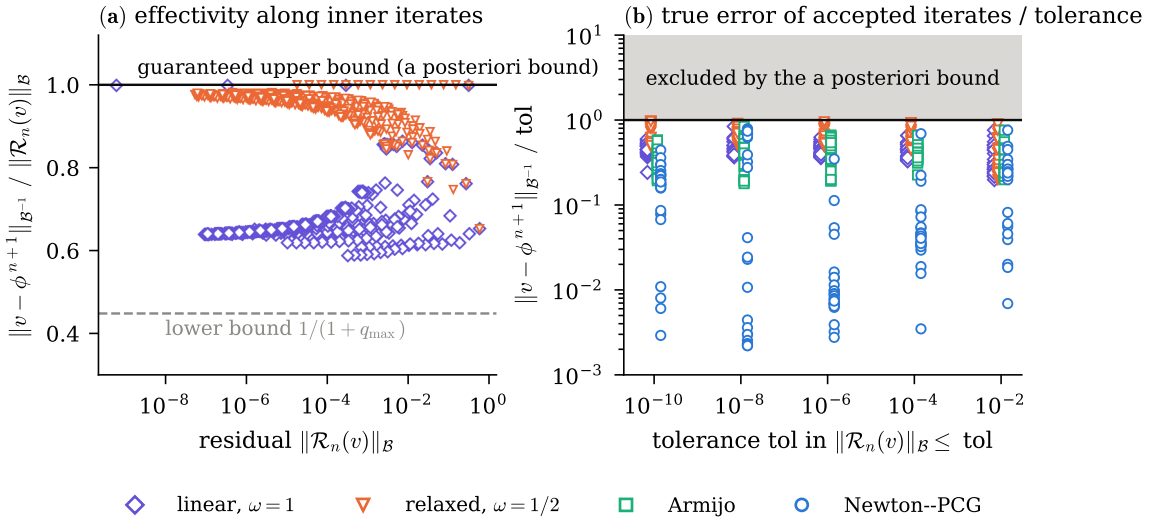

(a) effectivity along inner iterates
guaranteed upper bound (a posteriori bound)
lower bound $1/(1+q_{\max})$
$\|v-\phi^{n+1}\|_{\mathcal{B}^{-1}} \,/\, \|\mathcal{R}_n(v)\|_{\mathcal{B}^{-1}}$
1.0
0.8
0.6
0.4
10^{-8}
10^{-6}
10^{-4}
10^{-2}
10^{0}
residual $\|\mathcal{R}_n(v)\|_{\mathcal{B}}$
(b) true error of accepted iterates / tolerance
excluded by the a posteriori bound
$\|v-\phi^{n+1}\|_{\mathcal{B}^{-1}}$ / tol
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-10}
10^{-8}
10^{-6}
10^{-4}
10^{-2}
tolerance tol in $\|\mathcal{R}_n(v)\|_{\mathcal{B}} \leq$ tol
linear, $\omega = 1$
relaxed, $\omega = 1/2$
Armijo
Newton--PCG